DP Computer Science: Topic A4 Machine Learning — Assessment Markscheme

Overview

Total Marks: 40 marks

This markscheme covers the following topics:

- Machine Learning Fundamentals and Ethical Considerations
- Data Handling and Model Architectures
- Advanced Concepts and Evaluation

Question 1: Machine Learning Fundamentals and Ethical Considerations (13 marks)

(a) Machine Learning Types and Hardware [7 marks]

(i) Describe two distinct types of machine learning and provide real-world applications [4 marks]

Award: Up to 4 marks for describing two distinct ML types and providing a relevant, specific real-world application for each.

Supervised Learning:

Aspect	Description
Description	Training a model on a labelled dataset where input data is paired with correct outputs/labels. The model learns to map inputs to outputs.
Application	Medical diagnosis (predicting disease from patient symptoms and historical data), spam detection, image classification

Unsupervised Learning:

Aspect	Description
Description	Training on an unlabelled dataset where the model must find patterns, structures, or relationships independently.
Application	Customer segmentation (grouping customers by purchasing behaviour), anomaly detection, dimensionality reduction

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Reinforcement Learning (RL):

Aspect	Description
Description	An agent learns to make decisions by interacting with an environment, receiving rewards for desired actions and penalties for undesirable ones.
Application	Robotic cleaner (learning to navigate and clean through trial-and-error), game-playing AI (AlphaGo), autonomous navigation

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Deep Learning (DL):

Aspect	Description
Description	A subfield using artificial neural networks (ANNs) with multiple layers to learn complex patterns, effective for image and speech recognition.
Application	Image recognition (analysing chest X-rays for pneumonia), natural language processing, self-driving cars

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Transfer Learning (TL):

Aspect	Description
Description	A model developed for one task is reused as the starting point for a second, related task, reducing training time and data requirements.
Application	Adapting a pre-trained image recognition model to classify specific types of medical scans with a smaller dataset

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(ii) Explain the purpose of a GPU in deep learning training and describe one other specialised hardware component [3 marks]

Award: [1] for explaining GPU purpose, [1] for describing another hardware component, [1] for explaining how it aids deployment.

GPU (Graphics Processing Unit):

Aspect	Description
Primary Purpose	GPUs are crucial for training deep learning models because they are designed for **parallel processing**. Deep learning involves extensive matrix multiplications and convolutions, which can be performed much faster in parallel on a GPU compared to a traditional CPU, significantly accelerating training.

One other specialised hardware component (choose one):

Component	Description	Deployment Use
ASIC (Application-Specific Integrated Circuit)	A microchip designed for a specific application or function	Highly optimised for specific ML inference tasks, offering maximum speed and power efficiency

Component	Description	Deployment Use
FPGA (Field-Programmable Gate Array)	An integrated circuit that can be programmed or reconfigured after manufacturing	Offers a balance between flexibility and performance; can be updated post-deployment
TPU (Tensor Processing Unit)	A specialised AI accelerator developed by Google for neural network ML	Highly efficient for training and inference; provides very high performance for deep learning models
Edge Device	A device that performs computation at the "edge" of the network	Enables real-time inference and decision-making in scenarios where low latency and offline capability are critical
Cloud-based Platform	Remote servers providing computing services on demand	Offers scalability, flexibility, and cost-effectiveness; easy to scale resources based on demand
HPC Centre	Facilities housing supercomputers and computer clusters	Used for deploying highly complex or large-scale ML models requiring immense computational power

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(b) Ethical Implications of ML in Medical Diagnosis [6 marks]

Award: Up to 6 marks for a considered and balanced review of ethical implications, specifically focusing on algorithmic fairness and bias in medical diagnosis.

Discussion Points:

Point	Description
Algorithmic Bias and Fairness	Models learn from training data. If data reflects societal biases, the model can perpetuate and amplify them. A diagnostic model trained primarily on one ethnic group might perform poorly for underrepresented groups. This leads to unfair or inequitable outcomes in patient health and access to care.
Lack of Transparency	Deep learning models are often "black boxes". Understanding *why* a model arrived at a diagnosis is critical for trust and accountability. If a model is biased, lack of transparency makes it hard to identify and rectify the bias.
Accountability	When a model makes an incorrect diagnosis leading to harm, determining who is responsible (developer, hospital, physician) is complex. Clear lines of accountability are essential to address errors and prevent future occurrences.
Patient Autonomy and Informed Consent	Patients have the right to understand how their data is used and how diagnostic decisions are made. It is ethically imperative to obtain informed consent explaining the role of AI, its limitations (like bias), and the level of human oversight.
Privacy and Data Security	Medical data is highly sensitive. Protecting patient privacy and ensuring data security are paramount to prevent unauthorised access, data breaches, or unethical use of health information.
Reliability and Robustness	A model might fail in real-world scenarios due to overfitting or encountering unexpected data. Unreliable diagnostic tools can lead to misdiagnosis, patient harm, and erosion of trust in AI.

Question 2: Data Handling and Model Architectures (13 marks)

(a) Data Cleaning [4 marks]

(i) Explain the significance of data cleaning [2 marks]

Award: [1] for explaining what data cleaning addresses, [1] for explaining its impact on model performance.

Aspect	Description
Significance	Data cleaning addresses issues like inconsistent formats, missing values, incorrect data, irrelevant information, and duplicate entries. It is crucial because the quality of input data directly impacts model performance. "Garbage in, garbage out" applies: unclean data can lead to inaccurate predictions, biased results, and inefficient training.

(ii) Describe two techniques for handling missing data [2 marks]

Award: [1] for describing each distinct technique.

Technique	Description
Imputation	Filling in missing values with estimated values, such as the mean, median, or mode of the column, or using more advanced statistical methods or predictive models to infer the missing data.
Deletion	Removing rows (samples) that contain missing values, or removing entire columns (features) if they have a large proportion of missing data and are deemed less critical.
Predictive Modelling	Using other features in the dataset to predict and fill in the missing values for a specific feature.

Technique	Description
Flagging/Indicator Variable	Creating a new binary variable (0 or 1) that indicates whether a value was originally missing for a particular feature.

(b) Hyperparameter Tuning and Evaluation [5 marks]

(i) Outline the purpose of hyperparameter tuning [2 marks]

Award: [1] for stating the purpose, [1] for explaining how it improves model performance.

Aspect	Description
Purpose	Hyperparameter tuning is the process of selecting the optimal set of hyperparameters for a machine learning model. Its purpose is to optimise the model's performance (e.g., accuracy, generalisation ability) by finding the best configuration of these external parameters that govern the learning process. This helps prevent overfitting or underfitting.

(ii) Explain how the F1 score serves as a vital evaluation metric [3 marks]

Award: [1] for defining F1 score's purpose, [1] for explaining precision, [1] for explaining recall.

Aspect	Description
F1 Score	A vital evaluation metric, especially for imbalanced datasets (e.g., fraud detection). It provides a single score that balances **precision** and **recall**.
Precision	Measures the proportion of correctly identified positive cases out of all cases predicted as positive. High precision means that when the model predicts fraud, it is usually correct (minimises false alarms).

Aspect	Description
Recall (Sensitivity)	Measures the proportion of correctly identified positive cases out of all actual positive cases. High recall means the model successfully identifies most actual fraudulent transactions (minimises missed fraud).

(c) Artificial Neural Networks [4 marks]

(i) Sketch a simple Multi-Layer Perceptron (MLP) [2 marks]

Award: [1] for recognisable MLP structure, [1] for correctly labelling input, hidden, and output layers.

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│              MULTI-LAYER PERCEPTRON (MLP)                      │
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│    ┌─────────────┐    ┌─────────────┐    ┌─────────────┐      │
│    │   INPUT     │    │   HIDDEN    │    │   OUTPUT    │      │
│    │   LAYER     │    │   LAYER     │    │   LAYER     │      │
│    │             │    │             │    │             │      │
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│    Connections: Weighted connections (lines) link nodes from   │
│    one layer to the next, with arrows indicating information   │
│    flow.                                                        │
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(ii) Explain the primary function of hidden layers in an MLP [2 marks]

Award: [1] for stating the primary function, [1] for explaining how they derive patterns.

Aspect	Description
Primary Function	The primary function of hidden layers is to perform complex, non-linear transformations on the input data. They extract and combine features from the input, allowing the network to learn intricate and abstract patterns that are not immediately obvious in the raw input, enabling more sophisticated predictions or classifications.

Question 3: Advanced Concepts and Evaluation (14 marks)

(a) CNNs for Image Recognition [4 marks]

(i) Justify why a CNN is more appropriate than an MLP for image classification [4 marks]

Award: [1] for clear statement of why CNNs are better, up to [3] for specific justifications.

Justification	Description
Spatial Hierarchy and Feature Learning	CNNs are specifically designed to adaptively learn spatial hierarchies of features. Their convolutional layers automatically detect patterns like edges, textures, and shapes at various levels of abstraction without manual feature engineering. MLPs treat image pixels as a flat vector, losing crucial spatial relationships.
Weight Sharing/Parameter Efficiency	CNNs utilise weight sharing in their convolutional filters (same filter applied across different locations). This significantly reduces parameters compared to an MLP, making CNNs more efficient to train and less prone to overfitting.

Justification	Description
Translational Invariance	Pooling layers contribute to translational invariance—the network can still recognise an object even if its position shifts slightly. MLPs lack this capability.
Scalability	For high-resolution images common in autonomous vehicles, an MLP would require an enormous number of input neurons and connections, becoming computationally intractable. CNNs, with specialised layers and weight sharing, scale much more effectively.

(b) Overfitting and Underfitting [6 marks]

(i) Explain the concepts of overfitting and underfitting [4 marks]

Award: [1.5] for overfitting, [1.5] for underfitting, [1] for clearly distinguishing between them.

Concept	Description
Overfitting	Occurs when a model learns the training data too well, capturing noise and irregularities specific to the training set. The model performs exceptionally well on training data but poorly on unseen data (poor generalisation). It has memorised the training examples rather than learning generalisable rules.
Underfitting	Occurs when a model is too simple to capture the underlying relationships in the training data. The model performs poorly on both the training data and new data. It is unable to represent the complexity of the problem.
Distinction	An overfit model is **too complex** and excels on training data but fails on new data. An underfit model is **too simple** and fails on both training and new data.

(ii) Outline one practical strategy to mitigate overfitting [2 marks]

Award: [1] for identifying a strategy, [1] for outlining how it mitigates overfitting.

Strategy	Description
Increasing Training Data	Providing a larger and more diverse dataset helps the model learn more generalised patterns rather than memorising noise.
Regularisation (L1/L2, Dropout)	Adds a penalty to the loss function based on model complexity, discouraging large weights and simplifying the model. Dropout randomly deactivates neurons during training.
Early Stopping	Monitoring performance on a validation set and stopping training when performance degrades, even if training set performance is still improving.
Feature Reduction/Selection	Reducing the number of input features simplifies the model and reduces its ability to overfit to noisy or irrelevant data.
Cross-Validation	Using techniques like k-fold cross-validation helps estimate model performance on unseen data more robustly and detect overfitting.

(c) Model Selection and Comparison [4 marks]

(i) Evaluate the role and importance of model selection and comparison [4 marks]

Award: Up to 4 marks for evaluation (weighing strengths/limitations/importance) throughout the ML workflow.

Aspect	Description
Role and Importance	Model selection and comparison is a critical and iterative process throughout the entire ML workflow, ensuring the chosen model is the

Aspect	Description
	most appropriate and effective for the specific problem.
Initial Stages	Guides data collection and preprocessing; informs feature engineering and data splitting.
Training & Development	Train multiple candidate models (e.g., K-NN vs. Decision Trees). Use metrics (Accuracy, Precision, Recall, F1, R-squared) to objectively compare performance.
Strategic Considerations	Involves trade-offs: Performance vs. Complexity, Generalisation vs. Overfitting, Resource Allocation, and Ethical considerations (fairness across demographic groups).
Deployment & Post-Deployment	A well-selected model increases confidence in deployment. Models are continuously monitored and re-evaluated against new data.

Mark Allocation Summary

Question	Subpart	Marks
Q1	(a)(i)	4
	(a)(ii)	3
	(b)(i)	6
Q2	(a)(i)	2
	(a)(ii)	2
	(b)(i)	2

Question	Subpart	Marks
	(b)(ii)	3
	(c)(i)	2
	(c)(ii)	2
Q3	(a)(i)	4
	(b)(i)	4
	(b)(ii)	2
	(c)(i)	4
Total		**40**

End of Markscheme